Wenyou (Tobias) Tian

Year 4, Honours in Computer Science, Major in Economics
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Education

The University of British Columbia **Sep. 2022 – Jun. 2028**
Bachelor of Science, Honours in Computer Science *Vancouver, BC, Canada*
- Algorithm Design and Analysis; Software Engineering; Computer Systems; Computer Network; Quantum Computing

The University of British Columbia **Sep. 2022 – Jun. 2028**
Bachelor of Arts, Major in Economics *Vancouver, BC, Canada*
- Advanced Econometrics; Game Theory; Money and Banking; Micro/Macroeconomic Analysis; Intl. Finance/Trade

ETH Zürich - Department of Computer Science **Sep. 2024 – Feb. 2025**
Exchange Student *Zürich, Switzerland*
- Algorithms, Probability, Computing; Compiler Design; Computer Systems (OS + DS); Quantum Physics

Technical Skills

Computer Languages: Java, Python, C/C++, R, HTML/CSS/JavaScript, OCaml, TypeScript, SQL (basic)
Developer Tools: Git, VS Code, IntelliJ IDEA, JupyterLab, IDLE, Docker
Technologies/Frameworks: Matplotlib, pandas, NumPy, SymPy, Tidyverse, lubridate, haven, Node.js, Express
Linguistic Abilities: Mandarin (Native), English (C2), French (B1), German (A1)
Research Methodologies: OLS regression, Hypothesis Testing, RAG Pipelines, Agentic AI systems, Data Cleaning

Work Experience

Software Research Engineer **Jan. 2026 – Dec. 2026**
Noah's Ark Lab - Huawei Canada | Python, TypeScript, Git, Node.js, Agentic AI *Markham, Ontario*
- Rebuilt the training and hyperparameter-search pipeline for a phone screenshot popup detector fine-tuned from a YOLOv8 model that checks whether the phone application behaves according to UI/UX standards or not automatically; Fixed an incorrect optimizer fallback, reactivated a previously silent training mechanism, swapped out an outdated selection metric, and enforced multi-seed scoring between tuning hyperparameters and training the model; Ran 12 controlled experiments and curated a de-duplicated, leakage-free training set; Described efforts cut false popup detections by 85% (20 → 3 out of 262) and unresolved end-to-end cases by 40% (10 → 6 out of 90) with zero wrong verdicts, and improved accuracy from 82% to 95%.
- Led the design and implementation of a knowledge base harness, a TypeScript OpenCode plugin for querying large-scale knowledge bases, using a four-layer "deterministic kernel + LLM semantic agents" architecture aiming to process contents of at least 15,000 characters efficiently; Delivered a complete version which processes internal corporate design documents with a total content of 13,500+ characters, supporting 3 core functions of knowledge base initialization, query, and incremental update, aiding other developers to adhere to design guidelines and resolve bugs.
- Prototyped a tool for converting XML-based UI metadata into hierarchical UI trees, processing 700+ screenshots to support GUI agent interaction (project later transitioned across teams).

Undergraduate Academic Assistant **Jun. 2024 – Present**
Science One Physics, Year 1 Physics | Python, HTML/Markdown, Git, SymPy *The University of British Columbia*
- Reformatted and randomized more than 350 freshman-level physics problems with HTML and Markdown for frontend format display, and Python for backend question generation and randomization that assists 6+ physics professors to set up quizzes and exams, reaching hundreds of freshman students each year.
- Integrated the PrairieLearn autograder system to correctly identify and grade mathematical expressions that involve alphabetical variables and physical units using the SymPy library API which has served 151 students over two years.

Undergraduate Teaching Assistant (Mathematics and Physics) **Sep. 2023 – Apr. 2024**
Science One Program *The University of British Columbia*
- Held bi-weekly office hours and assisted in organizing tutorials and workshops that answered questions regarding freshman-level mathematics and physics, including calculus, classical and quantum mechanics, and special relativity.
- Coded short and simple Python programs to demonstrate mathematical and physical concepts numerically, such as Riemann sum approximation and Euler's method approximation.

Circulation and Shelving Assistant **Sep. 2023 – Apr. 2024**
Law Library *The University of British Columbia*
- Organized and rearranged books, journals, law statutes etc., by the Library of Congress Call Number System, across about 500 shelves each semester.
- Maintained book circulation within and across campuses, including assisting patrons with requested book information and personal inquiries, including answering phone calls.

Projects

Horizon Holdings | *Python, SQLite, Electron, Claude Code* **Aug. 2026**

- Specified and directed the build of a cross-platform portfolio tracker, whose code written entirely using Claude Code, modeling a self-designed 78-company, 12-market, 7-currency index against real brokerage holdings, in daily personal use for at least a month.
- Diagnosed and directed fixes for a currency-representation defect (LSE securities quote in GBX, not GBP) and a cross-timezone data-staleness gap between same-day US pricing and T+1 European/Japanese feeds.
- Set a data-integrity-first requirement where the system must refuse to record a portfolio value when any constituent price is missing, rather than defaulting silently

Insight UBC | *TypeScript, HTML/CSS/JavaScript, chai-as-promised, Mocha, Express, JSON, VS Code, Git* **Apr. 2025**

- Collaborated with a peer to develop a full-stack web application where users can upload and add, remove existing, list existing UBC's course section datasets, and perform queries on them in JSON-formatted files compressed as one zip file.
- Combined using chai-as-promised for smoke tests, unit tests, integration tests, and end-to-end tests that ensure the functionality of each data structure and the entire application.
- Utilized Express for the web application's framework to write RESTful APIs which allow sending requests and receiving responses with proper and reasonable status codes and results or errors.

Oat Compiler | *OCaml, VS Code, Docker, Linux* **Dec. 2024**

- Developed a complete compiler for the Oat language, encompassing an x86lite simulator, LLVMlite-based IR construction, type checking, and simple optimizations over a four-month period.
- Designed and implemented a register allocator using Kempe's K-coloring algorithm to optimize register utilization compared to the previous spill-to-stack approach.
- Implemented core Oat language features, including arrays, functions, and control flow constructs, while validating correctness through systematic testing of each compilation phase.

Classical Enigma Machine | *Java, JUnit, JSON, Swing, IntelliJ IDEA* **Dec. 2023**

- Digitalized a German cryptographic device used during WWII, and expanded the functionality to incorporate more input/output characters subject to user customization.
- Utilized JUnit testing library to ensure the correctness of each class method, and guaranteed data persistence with JSON for users to load or save at each session.
- Designed a GUI using the Swing library in which users can interact with the functions of the program.

Papers and Publications

Quantum Monte Carlo | *Quantum Mechaniscs, Quantum Computing, Python, JupyterLab* **Dec. 2025**

- Co-authored an expository paper discussing and explaining how three types of Monte Carlo methods can be utilized to simulate and derive variables of interest in a physical setting.
- Built upon quantum mechanics and quantum computing concepts to conceptualize and provided a mathematical interpretation of the quantum environment.

A Research on UBC Dining Halls' Food Waste | *R, Tidyverse, Econometric Analysis* **Apr. 2024**

- Applied multivariate OLS regression to 270+ observations across three dining halls to quantify the impact of meal-period foot traffic on daily food waste.
- Diagnosed heteroskedasticity using White's test and re-estimated models with heteroskedasticity-consistent (HC) robust standard errors; performed hypothesis testing, including F-tests, by meal period.
- Identified statistically significant effects at Orchard Commons Dining Hall (breakfast $\beta = -0.25$, lunch $\beta = +0.37$, both $p < 0.01$) and validated results through robustness checks using simplified model specifications.

On Hayek: Interpretation of Spontaneous Order | *Economics, Philosophy, Logic* **Apr. 2024**

- Compared and contrasted opinions from two groups of scholars debating whether or not Hayek committed naturalistic fallacy in his argument for a spontaneous order.
- Analyzed and identified flaws in each side's argument and proposed more accurate interpretations of Hayek's work and literature.
- Concluded that each side has their weaknesses in their claim, however both agreed upon that Hayek did not commit naturalistic fallacy in his economic worldview.

Earth's Albedo via Relative Photometry | *Python, NumPy, Matplotlib, Mathematical modelling* **Apr. 2023**

- Derived and implemented a photometric Earth albedo model from primary literature, incorporating earthshine ratios, lunar phase functions, and atmospheric extinction corrections.
- Processed telescope image data using AstroImageJ and performed Python-based analyses across ISO and exposure parameter space; estimated Earth albedo at 0.262 (vs. literature value of 0.30).
- Presented results at a student-led academic conference and co-authored a 29-page research paper with a faculty mentor in the Department of Physics and Astronomy at UBC.

Scholarships and Awards

J Fred Muir Memorial Scholarship in Science **Mar. 2026**

Value: CA$ 210 *The University of British Columbia*

- Awarded to students in the Faculty of Science on the recommendation of the Faculty for academic excellence.

Hugh M Brock Go Global Award **Sep. 2024**

Value: CA$ 1,500 *The University of British Columbia*

- Award to UBC students participating in international learning programs through Go Global.